Project 7:-Maven Selenium Project – Automating Web Applications Using Firefox Browser

Objective

The objective is to build a Maven-based Selenium automation script that uses Firefox to perform login, product search, and authentication across three different websites in separate browser tabs.

- Login to the SauceDemo website
- Search and add a product to the cart on AutomationExercise website
- Login to the Practice Test Automation website
- Demonstrate **multi-tab browser automation using Selenium WebDriver**

Websites Used for Automation

Automation is performed on the three websites listed below:

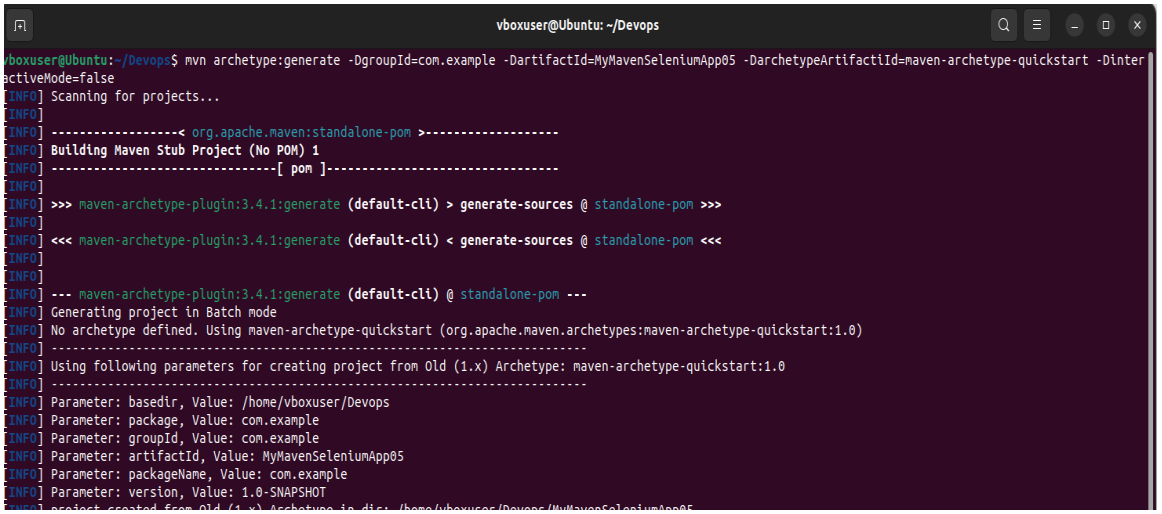
1. SauceDemo
   https://www.saucedemo.com/
2. Automation Exercise
   https://automationexercise.com/products
3. Practice Test Automation
   https://practicetestautomation.com/practice-test-login/

Step 1: Create a Maven Project

Use the following terminal command to create the Maven project structure:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenFirefoxApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

Project Structure:



```
vboxuser@Ubuntu:~/Desktop/Devops/Lab2/MyMavenSeleniumApp05$ tree
.
├── dependency-reduced-pom.xml
├── pom.xml
├── src
│   ├── main
│   │   └── java
│   │       └── com
│   │           └── example
│   │               └── App.java
│   └── test
│       └── java
│           └── com
│               └── example
│                   └── AppTest.java
└── target
    ├── classes
    │   └── com
    │       └── example
    │           └── App.class
    ├── generated-sources
    │   └── annotations
    ├── generated-test-sources
    │   └── test-annotations
    ├── maven-archiver
    │   └── pom.properties
    ├── maven-status
    │   └── maven-compiler-plugin
    │       ├── compile
    │       │   └── default-compile
    │       │       ├── createdFiles.lst
    │       │       └── inputFiles.lst
    │       └── testCompile
    │           └── default-testCompile
    │               ├── createdFiles.lst
    │               └── inputFiles.lst
```

Step 2: Install Firefox Browser

First verify that Firefox is available on your system:

firefox --version

Example output:

Mozilla Firefox 124.0

If the above command fails, install Firefox using:

sudo apt install firefox

Step 3: Install GeckoDriver

GeckoDriver acts as a bridge between Selenium and Firefox. Follow these steps to install it on your system:

Download GeckoDriver:

wget https://github.com/mozilla/geckodriver/releases/latest/download/geckodriver-v0.34.0-linux64.tar.gz

Extract the file:

tar -xvzf geckodriver-v0.34.0-linux64.tar.gz

Move the binary to a location on the system PATH:

sudo mv geckodriver /usr/local/bin/

sudo chmod +x /usr/local/bin/geckodriver

Verify installation:

geckodriver --version

Step 4: Add Selenium Dependency in pom.xml

Edit the pom.xml file and insert the following Selenium dependency inside the <dependencies> section:

pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
        http://maven.apache.org/maven-v4_0_0.xsd">

<modelVersion>4.0.0</modelVersion>

<groupId>com.example</groupId>
<artifactId>MyMavenFirefoxApp</artifactId>
<version>1.0-SNAPSHOT</version>
<packaging>jar</packaging>


<name>MyMavenFirefoxApp</name>
<dependencies>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
</dependencies>
<build>
<plugins>
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.11.0</version>
<configuration>
<source>11</source>
<target>11</target>
</configuration>
</plugin>
```

```
<plugin>
<groupId>org.codehaus.mojo</groupId>
<artifactId>exec-maven-plugin</artifactId>
<version>3.1.0</version>
<configuration>
<mainClass>com.example.App</mainClass>
</configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 5: Inspect Elements in the Websites

Before coding, use the browser's Inspect tool to identify the HTML elements that will be automated.

Steps:

1. Visit the target website in your browser.
2. Right-click on the page to access the context menu.
3. Click Inspect to open the browser developer tools.
4. Use the element picker in DevTools to hover over and select individual elements on the page.

Important IDs used in this project:

SauceDemo

Field	**ID**
Username	user-name
Password	password
Login Button	login-button

Automation Exercise

Field	ID
Search Box	search_product
Search Button	submit_search
Product Button	data-product-id

Practice Test Automation

Field	ID
Username	username
Password	password
Login Button	submit

Step 6: Create App.java

Location:

src/main/java/com/example/App.java

Java Code:

```
package com.example;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.WindowType;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import java.time.Duration;

public class App {

    public static void main(String[] args) throws InterruptedException {

        FirefoxOptions options = new FirefoxOptions();

        WebDriver driver = new FirefoxDriver(options);

        driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920,1080));

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));

        Actions actions = new Actions(driver);
```

```
driver.get("https://www.saucedemo.com/");

driver.findElement(By.id("user-name")).sendKeys("standard_user");

driver.findElement(By.id("password")).sendKeys("secret_sauce");

driver.findElement(By.id("login-button")).click();

System.out.println("SauceDemo login successful");

driver.switchTo().newWindow(WindowType.TAB);

driver.get("https://automationexercise.com/products");

driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");

driver.findElement(By.id("submit_search")).click();

WebElement product = wait.until(
    ExpectedConditions.visibilityOfElementLocated(
        By.cssSelector("a[data-product-id='2']")
    )
);

((org.openqa.selenium.JavascriptExecutor) driver)
    .executeScript("arguments[0].scrollIntoView({block:'center'});", product);

((org.openqa.selenium.JavascriptExecutor) driver)
    .executeScript("arguments[0].click();", product);

WebElement viewCart = wait.until(
    ExpectedConditions.elementToBeClickable(
        By.cssSelector("#cartModal a[href='/view_cart']")
    )
);

((org.openqa.selenium.JavascriptExecutor) driver)
    .executeScript("arguments[0].scrollIntoView(true);", viewCart);

((org.openqa.selenium.JavascriptExecutor) driver)
    .executeScript("arguments[0].click();", viewCart);

System.out.println("Automation Exercise product added to cart");

driver.switchTo().newWindow(WindowType.TAB);

driver.get("https://practicetestautomation.com/practice-test-login/");

Thread.sleep(2000);

driver.findElement(By.id("username")).sendKeys("student");

driver.findElement(By.id("password")).sendKeys("Password123");
```

```
        driver.findElement(By.id("submit")).click();

        System.out.println("Practice Test Automation login successful");

        Thread.sleep(5000);

        driver.quit();
    }
}
```

Open *App.java ~/DevOps/MyMavenFirefoxApp/src/main/java/com/example Save

```
1 package com.example;
2
3 import org.openqa.selenium.By;
4 import org.openqa.selenium.WebDriver;
5 import org.openqa.selenium.WebElement;
6 import org.openqa.selenium.WindowType;
7 import org.openqa.selenium.firefox.FirefoxDriver;
8 import org.openqa.selenium.firefox.FirefoxOptions;
9 import org.openqa.selenium.interactions.Actions;
10 import org.openqa.selenium.support.ui.ExpectedConditions;
11 import org.openqa.selenium.support.ui.WebDriverWait;
12 import java.time.Duration;
13
14 public class App {
15
16     public static void main(String[] args) throws InterruptedException {
17
18         FirefoxOptions options = new FirefoxOptions();
19         WebDriver driver = new FirefoxDriver(options);
20         driver.manage().window().setSize(new org.openqa.selenium.Dimension(1920, 1080));
21
22         WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(15));
23         Actions actions = new Actions(driver);
24
25         driver.get("https://www.saucedemo.com/");
26         driver.findElement(By.id("user-name")).sendKeys("standard_user");
27         driver.findElement(By.id("password")).sendKeys("secret_sauce");
28         driver.findElement(By.id("login-button")).click();
29         System.out.println("SauceDemo login successful");
30
31         driver.switchTo().newWindow(WindowType.TAB);
32         driver.get("https://automationexercise.com/products");
33         driver.findElement(By.id("search_product")).sendKeys("Men Tshirt");
34         driver.findElement(By.id("submit_search")).click();
35
36         WebElement product = wait.until(
37                 ExpectedConditions.visibilityOfElementLocated(By.cssSelector("a[data-product-id='2']"))
38         );
39
40         ((org.openqa.selenium.JavascriptExecutor) driver)
41                 .executeScript("arguments[0].scrollIntoView({block:'center'});", product);
42         ((org.openqa.selenium.JavascriptExecutor) driver)
43                 .executeScript("arguments[0].click();", product);
44
```

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```
45         WebElement viewCart = wait.until(
46                 ExpectedConditions.elementToBeClickable(By.cssSelector("#cartModal a[href='/view_cart']"))
47         );
48         ((org.openqa.selenium.JavascriptExecutor) driver)
49                 .executeScript("arguments[0].scrollIntoView(true);", viewCart);
50         ((org.openqa.selenium.JavascriptExecutor) driver)
51                 .executeScript("arguments[0].click();", viewCart);
52
53         System.out.println("Automation Exercise product added to cart");
54
55         driver.switchTo().newWindow(WindowType.TAB);
56         driver.get("https://practicetestautomation.com/practice-test-login/");
57         Thread.sleep(2000);
58         driver.findElement(By.id("username")).sendKeys("student");
59         driver.findElement(By.id("password")).sendKeys("Password123");
60         driver.findElement(By.id("submit")).click();
61         System.out.println("Practice Test Automation login successful");
62
63         Thread.sleep(5000);
64         driver.quit();
65     }
66 }
```

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Step 7: Build the Maven Project

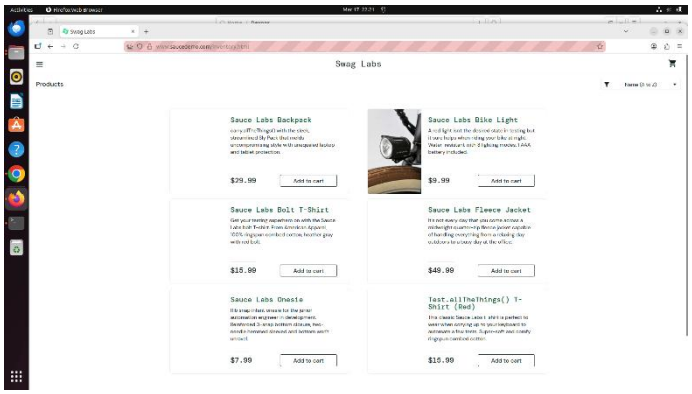
Run the command below to compile the code and build the project:

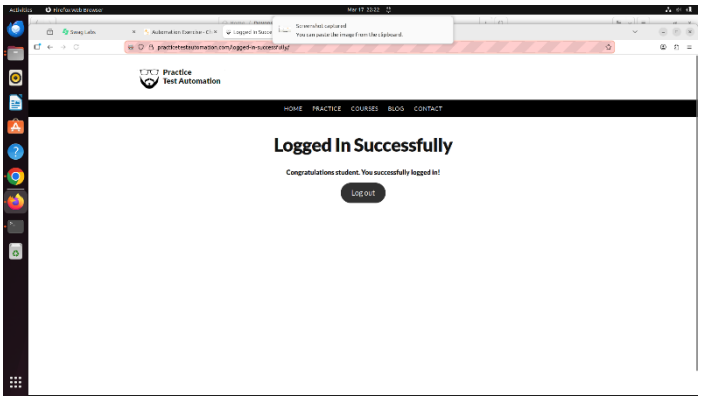
mvn clean compile

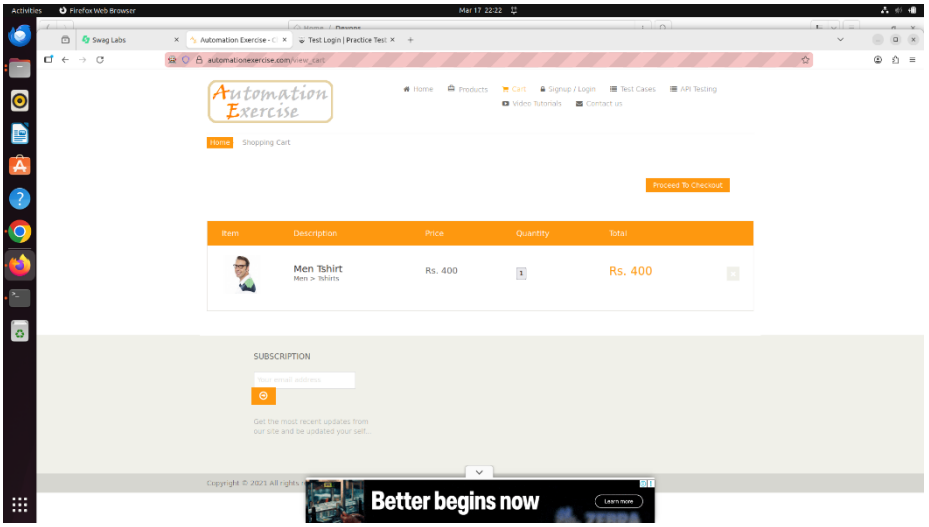
Step 8: Run the Application

Once the build is successful, launch the application with:

mvn exec:java -Dexec.mainClass="com.example.App"







Name : Someesvaar V
USN : 1BI23CS210